1.

Full data set

The adjacent table gives the number of worldwide Internet users, in millions.

Complete parts **(a)** through **(b)**.

Year	Users (millions)	Year	Users (millions)
1995	17	2002	597
1996	34	2003	721
1997	74	2004	819
1998	151	2005	1025
1999	245	2006	1101
2000	367	2007	1229
2001	539		

(a) Find the cubic function that is the best fit for the data, with y equal to the number of millions of users and x equal to the number of years from 1990.

y = ☐

(Use integers or decimals for any numbers in the expression. Round to three decimal places as needed.)

(b) Use the model to predict the number of users in 2013.

The number of users in 2013 will be approximately ☐ million.

(Round to the nearest whole number as needed.)

2. The size of the civilian labor force (in millions) for selected years from 1950 and projected to 2050 can be modeled by $y = -0.0001x^3 + 0.0058x^2 + 1.43x + 56.7$, where x is the number of years after 1950.

(a) What does this model project the civilian labor force to be in 2013?

(b) In what year does the model project the civilian labor force to be 152 million?

(a) In 2013 the model projects the civilian labor force to be ☐ million.

(Type an integer or decimal rounded to one decimal place as needed.)

(b) The model projects the civilian labor force to be 152 million in ☐

(Round up to the next year.)

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3. The weekly revenue for a product is given by $R(x) = 97.2x - 0.0135x^2$, and the weekly cost is $C(x) = 9000 + 48.6x - 0.027x^2 + 0.00001x^3$, where x is the number of units produced and sold.

(a) How many units will give the maximum profit?

(b) What is the maximum possible profit?

(a) The number of units that will give the maximum profit is ☐.

(Round to the nearest whole number as needed.)

(b) The maximum possible profit is $☐.